

Pre-Lab 1: Environment Setup

Getting Ready for the First In-Class Lab

Course: Machine Learning for Single-Cell and Spatial Biology (Prof. Uthsav Chitra, JHU)

Unit 1 — Linear Dimensionality Reduction | Prepared by Sai Dharmasena

Overview

This pre-lab ensures you have the tools and environment ready **before coming to class**. You will set up Python, Git, an AI coding assistant, and a GitHub account, then clone the course starter repo and verify everything works. Completing this pre-lab means you can spend class time on the actual lab, not on installation.

Estimated time: 30–45 minutes. **Due:** Before Class 2 (math review). **Deliverable:** Submit your GitHub repo link on Canvas.

Step 1: Install Python & Git

- Install **Python 3.9+** from python.org (check with `python --version`)
- Install **Git** from git-scm.com/downloads (check with `git --version`)
- Both are free and available on Mac, Windows, and Linux

Step 2: Install an AI Coding Assistant

We will use AI agents throughout the course to help with data analysis. Install at least one:

- **Claude Code:** `npm install -g @anthropic-ai/claude-code` (requires Node.js)
- **Codex:** see OpenAI documentation for installation
- Or use any AI coding assistant you're comfortable with (Cursor, GitHub Copilot, etc.)

You don't need to be an expert with these tools yet — we'll learn together in class. The goal is to have something installed and accessible before the first lab.

Step 3: Create a GitHub Account

- Go to github.com and sign up (free)
- Set your Git identity so commits are attributed to you:

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Step 4: Clone the Course Starter Repo

Your instructor will provide a GitHub URL for the course starter repo. Clone it to your machine:

```
git clone  
cd starter_repo
```

The repo contains:

File	Purpose
README.md	This guide + course info
hello_lab.ipynb	Notebook to verify your setup works
test_setup.py	Script that checks your environment
.gitignore	Tells Git which files to ignore

Step 5: Install Python Packages

Install the core packages we'll use throughout the course:

```
pip install numpy pandas matplotlib scikit-learn scipy jupyter
```

If you use conda, the equivalent is:

```
conda install numpy pandas matplotlib scikit-learn scipy jupyter
```

Step 6: Run the Environment Check

Run the setup verification script:

```
python test_setup.py
```

You should see a series of **[PASS]** messages in green. If any check **[FAIL]**s, the script prints a fix hint — follow it. Common issues:

- Missing package → run `pip install <package>`
- Git not configured → run the `git config` commands from Step 3
- Python version too old → install Python 3.9+
- Jupyter not found → run `pip install jupyter`

Step 7: Run the Hello Notebook

Launch Jupyter and open the hello notebook:

```
jupyter notebook hello_lab.ipynb
```

Run all cells from top to bottom (Shift+Enter to run each cell). You should see:

- Printed version numbers for NumPy, Pandas, Matplotlib
- A **PCA scatter plot** with 3 clearly separated colored clusters
- A green **"You're ready for Lab 1!"** message at the bottom
- A file `hello_lab_pca.png` saved in the repo folder

Step 8: Commit & Push to GitHub

Save your work and push it to your GitHub repo:

```
git add .
git commit -m "Completed pre-lab 1 setup"
git push
```

This creates a record that you completed the pre-lab. The generated plot (`hello_lab_pca.png`) will be included in your commit.

Step 9: Submit Your GitHub Link

- Go to your repo on github.com
- Copy the URL from your browser's address bar
- Submit it on Canvas under the "Pre-Lab 1" assignment

Troubleshooting

Problem	Solution
<code>`python`</code> command not found	On Mac/Linux, try <code>`python3`</code> . On Windows, reinstall Python and check 'Add to PATH'
<code>`pip`</code> command not found	Try <code>`python -m pip install`</code> instead
<code>`git clone`</code> fails	Check the URL. Make sure you're logged into GitHub. Try HTTPS URL if SSH fails
<code>`git push`</code> asks for password	Set up a Personal Access Token (GitHub Settings → Developer settings → Tokens)
Jupyter won't start	Run <code>`pip install jupyter`</code> again. Try <code>`jupyter lab`</code> as an alternative
Notebook cells error	Make sure you ran <code>`pip install`</code> for all packages. Restart the kernel and try again
AI assistant not working	Not required for pre-lab. We'll set it up together in class if needed

Tip: If you get stuck, use your AI coding assistant to help debug. Paste the error message and ask "How do I fix this?". That's exactly how we'll use these tools in the labs!

Pre-Lab Checklist

Use this checklist to make sure you've completed everything before class:

#	Task	Done?
1	Python 3.9+ installed and verified	■
2	Git installed and verified	■
3	AI coding assistant installed (Claude Code, Codex, etc.)	■
4	GitHub account created	■
5	Git user.name and user.email configured	■
6	Course starter repo cloned	■
7	Python packages installed (numpy, pandas, etc.)	■
8	test_setup.py — all checks pass	■
9	hello_lab.ipynb — all cells run successfully	■
10	PCA plot generated and visible	■
11	Changes committed and pushed to GitHub	■
12	GitHub repo link submitted on Canvas	■

Notes

- **This pre-lab ties into Class 2 (math review).** We'll use the setup you complete here during the in-class lab.
- **The AI assistant is recommended but not required** for the pre-lab. We'll integrate it more in the labs themselves.
- **If you're new to Git/GitHub**, don't worry — the workflow is simple: clone, commit, push. We'll practice this every lab.
- **If you're new to Python/Jupyter**, the hello notebook is intentionally simple. The labs will get more interesting.
- **Having an AI assistant installed** before class means you can follow along when we demonstrate using agents for data analysis.
- **The starter repo is your personal workspace.** You'll add lab notebooks, figures, and reflections to it throughout the course.